

Semiconductor Device Simulator

Basic Theory

Copyright ©Qiusong Chen

Version: 1.0

November 20, 2023

Contents

1	Objectives and Background	1
1.1	Software Component Modules	1
1.2	Installation Procedures	2
2	Basic Physical Theory of Devices	4
2.1	Crystalline Semiconductor Theory	4
2.1.1	Energy-Band Structure	4
2.1.2	Fermi-Dirac Distribution Statistics	5
2.1.3	Definition of Electrons and Holes	6
2.1.4	Boltzmann Distribution Approximation	6
2.1.5	Carrier Concentration	7
2.1.6	Quasi-Fermi Levels	8
2.1.7	Mobility	8
2.1.8	Einstein Relation	9
2.1.9	Current Density	9
2.2	Theoretical Analysis of Amorphous Semiconductors	9
2.2.1	Energy-State Structure	10
2.2.2	Carrier Concentration	11
2.2.3	Quasi-Fermi Levels	12
2.2.4	Mobility	12
2.2.5	Generalized Einstein Relation	13
2.3	Analytical Methods for Defect States in Organic Semiconductors	14
2.3.1	Gaussian and Exponential Distributions	14
2.3.2	Distribution Functions and Occupation of Donor-Like Trap States	15
2.3.3	Miller-Abrahams Formula	15
2.3.4	Trap-Charge Generation Process	16
2.3.5	Trap-Charge Recombination Process	17
2.3.6	Effective Unoccupied and Occupied Densities	18
2.3.7	Net Generation Rate for Trap Charges	19
2.3.8	Conversion between Areal and Volume Trap Densities	19
2.4	Basic Theory of Ferroelectric Materials	20
2.4.1	Landau-Ginsberg Theory	20
2.4.2	Miller Empirical Formula	21
3	Numerical Simulation Methods for Semiconductor-Ferroelectric Devices	23
3.1	Core Concepts of Semiconductor Device Simulation	23
3.1.1	Mesh Discretization	24
3.1.2	Finite-Volume Discretization	25
3.2	Conventional Flux Evaluation in Finite-Volume Method	25
3.3	Flux Evaluation Using Element-Edge Finite-Volume Method	26
3.3.1	Definition of Element Edge	26

3.3.2	Assembly of Discrete Equations for Newton-Raphson Solver	28
3.3.3	Gradient-Vector Evaluation on Element-Edges	28
3.3.4	Practical Remarks on Element-Edge Computation	30
3.4	Assembly of Semiconductor Device Equations	31
3.4.1	Electron and Hole Continuity Equations	31
3.4.2	Gauss' s Law for Electrostatics	31
3.4.3	Scharfetter-Gummel Discretization	32
3.4.4	Semiconductor-Electrode Interfaces —Boundary Conditions	33

1 Objectives and Background

This software adopts numerical computation methods to simulate electronic devices with physical properties such as semiconductors and ferroelectrics. Simulation results are used to analyze internal physical processes of devices, including current-voltage characteristics, generation-recombination processes of semiconductor defect states, and polarization processes inside ferroelectric materials.

The core solving engine of this software is DEVSIM, an open-source semiconductor device simulator. This software is secondary-developed based on DEVSIM and provides dedicated solutions for specific types of physical devices.

DEVSIM constructs iterative formulas for Gauss' s law of the electric field and carrier continuity equations in semiconductors using the finite-volume method, and solves them with the Newton-Raphson iteration. Its iterative formulas are mainly built via symbolic functions. Therefore, the formulas for various physical models of electronic devices can be conveniently implemented in corresponding symbolic-function form, which greatly facilitates the implementation and solution of complex physical models. Understanding the basic concepts and usage of DEVSIM is a prerequisite for using this software. For detailed mathematical principles and user documentation of DEVSIM, please refer to its website <https://devsim.net/>.

1.1 Software Component Modules

This software currently contains the following functional modules. Each module is implemented as a Python extension library:

1. Create, delete, and verify DEVSIM models
QSmodel_create.py
2. Build physical-relationship models for silicon semiconductors
QSimple_physics.py
3. Build physical-relationship models for defect states
QSExtraCharge.py
4. Build physical-relationship models for ferroelectric materials
QSFerro.py
5. Perform device voltage sweep
QSPlotSweep.py
6. Build physical-relationship models for Schottky contacts
QSSchottkyContact.py
7. Construct one-dimensional and two-dimensional diode simulation devices
QSDiode_common.py

1.2 Installation Procedures

This software shares exactly the same supported platforms as DEVSIM. Python serves as the execution backend, so it can run on Linux, macOS, and Windows operating systems. Conda or MiniConda under Linux is recommended for Python environment configuration.

Apart from DEVSIM, the following open-source software packages are required for pre-processing and post-processing workflows:

1. Multi-dimensional mesh generator: Gmsh <https://gmsh.info/>
2. Data visualization and figure-generation software: VisIt <https://www.visitusers.org/>

The installation steps below are demonstrated for the Ubuntu Linux operating system with MiniConda:

1. Install MiniConda

Download and install MiniConda for your hardware platform following instructions on the [MiniConda webpage](#), or install directly via the command-line.

2. Install DEVSIM

Download and install DEVSIM for your hardware platform following instructions on the [DEVSIM webpage](#).

3. Install Gmsh

Run the following command in the Linux terminal to install Gmsh:

```
sudo apt install gmsh
```

4. Install VisIt

Download and install VisIt for your hardware platform according to the guidance on the VisIt webpage <https://www.visitusers.org/>.

5. Required Python extension libraries

Essential Python packages for this software include: mkl, numpy, matplotlib, pandas, scipy, sympy, etc. Run the following command in the Linux terminal for installation:

```
conda install mkl numpy matplotlib pandas scipy sympy
```

6. Use the software modules

Place all Python extension library files under the same directory. Import modules using the Python statements shown below:

```
from devsim import *  
import sys, devsim  
sys.path.append('[Your library storage path]')  
from [module_name] import *
```

2 Basic Physical Theory of Devices

2.1 Crystalline Semiconductor Theory

For the convenience of constructing mathematical models, solid-state physics mainly builds the microscopic theoretical framework of matter based on crystalline materials with periodically arranged atoms, i.e., taking crystals as the fundamental research object. Besides facilitating mathematical modeling, crystal structures also allow probing their microstructures with high-energy particles. The most common techniques include X-ray diffraction, neutron diffraction, and electron diffraction. The research environment combining theory and experiment provides a powerful analytical platform for solid-state physics, enabling substantial progress in understanding the intrinsic physical properties of crystalline materials and laying a solid foundation for the subsequent emergence of semiconductor technology.

Two widely-used models describing electron motion in crystals are the nearly-free-electron model and the tight-binding model. Both are grounded in quantum mechanics [1] and solve the wave functions for electrons moving around periodically arranged ion cores. The difference lies in their zero-order approximations: the nearly-free-electron model adopts wave functions of free electrons in an average potential field as the zero-order approximation and treats the non-average Coulomb potential of periodically distributed ion cores as a perturbation. In contrast, the tight-binding model uses wave functions of outer-shell atomic electrons as the zero-order approximation and treats the Coulomb potential from neighboring ion cores as a perturbation. Despite their different treatments of the ion-core Coulomb potential, both models arrive at the same conclusion: electronic energy states in a periodic potential form band-like distributions known as energy bands, where electrons exist in a crystal-delocalized form. Energy bands permitting electron occupation are called allowed bands; energy ranges between allowed bands with no available electronic states are band gaps (forbidden bands). According to electron occupation, allowed bands are classified into the conduction band (the lowest unoccupied band) and the valence band (the highest occupied band). Based on the band structure and electron filling status, solid materials are categorized into conductors, insulators, and semiconductors [2].

Although the fundamental definitions of semiconductors originate from crystalline structures, these concepts can be generalized to non-crystalline materials to reveal the physical origins of various solid-state properties and establish a new analytical framework for electron dynamics.

2.1.1 Energy-Band Structure

In a crystal, microscopic constituents such as atoms, ions, or molecules arrange periodically at lattice sites to reduce binding energy and vibrate around these positions under thermal effects. These positions are lattice points, and the collection of lattice points forms the crystal lattice [1, 2]. Specific bonding configurations vary for different particles, transforming originally isolated atomic quantum states into delocalized wave functions shared across the entire lattice with discrete energy levels. Owing to the enormous number of lattice sites, the spacing between discrete energy levels becomes extremely small, eventually forming band-shaped energy-state distributions, as illustrated in Figure 1.

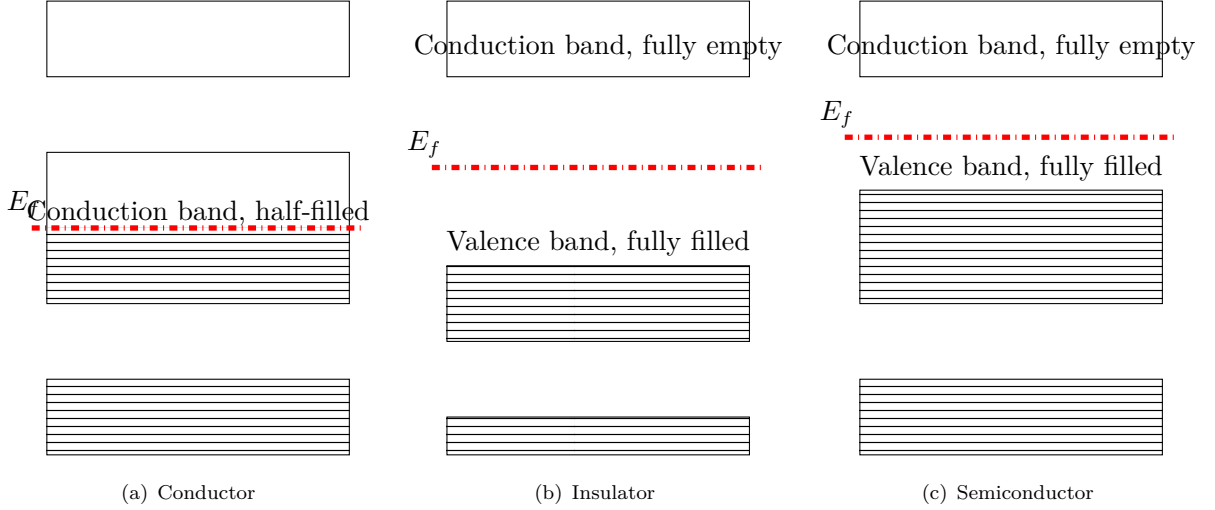


Figure 1: Schematic diagram of crystal energy-level structure and Fermi level

Energy levels with available electronic states are called **allowed bands**; energy ranges without electronic states are **band gaps**. According to actual electron occupation, crystals fall into three categories:

- **Conductor:** A material is a conductor when the Fermi level lies inside the conduction band (Figure 1(a)). Empty states exist near the Fermi level within the conduction band, allowing electrons to readily transit and yielding good electrical conductivity.
- **Insulator:** In insulators, the Fermi level sits between the conduction band and valence band (Figure 1(b)). The band-gap width is large, so electrons can hardly jump across the gap and remain confined in their original states, resulting in negligible conductivity.
- **Semiconductor:** Semiconductors possess band structures similar to insulators, with the Fermi level located within the band gap (Figure 1(c)). Nevertheless, their band-gap width is small; under appropriate excitation, electrons from the valence band can easily be excited into the conduction band.

2.1.2 Fermi-Dirac Distribution Statistics

Once a crystal structure is fixed, the set of accessible electronic states is determined. Electrons are fermions: each quantum state holds at most one electron, or two electrons when spin degeneracy is considered. The probability that a particular state is occupied by an electron is governed by the Fermi-Dirac distribution function:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_f}{k_B T}\right)} \quad (1)$$

where $f(E)$ denotes the occupation probability of a state with energy E ; E_f is the Fermi level (chemical potential) characterizing the electron-distribution of the system, shown as the red line in Figure 1; k_B is the Boltzmann constant; and T is the thermodynamic temperature of

the system. As indicated by the formula, occupation probabilities depend on temperature, so electronic transitions often exhibit thermally activated behavior. To position the Fermi level relative to band edges for semiconducting behavior, suitable crystalline materials combined with precise doping techniques are required.

2.1.3 Definition of Electrons and Holes

Owing to fermionic statistics, electrons can only transition from occupied states to unoccupied states. To satisfy conservation laws, electrons absorb or release energy and momentum to the lattice via photons and phonons; intra-band transitions are therefore more probable. At absolute zero, the conduction band of a semiconductor contains no electrons, while the valence band is completely filled. As temperature rises, thermal excitation promotes some electrons from the top of the valence band across the band gap into the conduction band. These excited electrons can move freely near the conduction-band minimum and produce current under electric-field or concentration-gradient driving forces. They retain electronic properties including charge, momentum, and spin and are referred to as electrons.

When electrons are thermally excited out of valence-band states, positively charged ion cores and empty quantum states remain. Electrons in neighboring energy ranges can move under electric-field or concentration-gradient stimuli, manifesting macroscopically as mobile positive vacancies. These mobile positively charged vacancies are defined as holes.

According to these definitions, semiconductors contain at least two charge carriers, collectively termed carriers.

2.1.4 Boltzmann Distribution Approximation

Although electron occupation in allowed bands follows the Fermi-Dirac distribution, direct evaluation of this formula is computationally intractable in practical device simulation. Simplifications are introduced for semiconductor conditions. At room temperature of 300 K, the thermal energy $k_B T$ equals 25.852 meV, which is much smaller than typical semiconductor band gaps. For any state in the conduction band, $E - E_f \gg k_B T$, so Equation (1) reduces to:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_f}{k_B T}\right)} \approx \exp\left(-\frac{E - E_f}{k_B T}\right) \quad (2)$$

This form matches the Boltzmann distribution and is known as the Boltzmann approximation.

For holes in the valence band, we evaluate the probability that a state is unoccupied. Since $E_f - E \gg k_B T$ near the valence-band top (Figure 1(c)), the hole occupation probability becomes:

$$\begin{aligned} 1 - f(E) &= 1 - \frac{1}{1 + \exp\left(\frac{E - E_f}{k_B T}\right)} \\ &= \frac{1}{1 + \exp\left(\frac{E_f - E}{k_B T}\right)} \\ &\approx \exp\left(-\frac{E_f - E}{k_B T}\right) \end{aligned} \quad (3)$$

2.1.5 Carrier Concentration

Given the density of states of allowed bands, carrier concentrations under thermal equilibrium can be computed from the distribution functions. While electron energies are discrete due to band gaps, electron distributions in momentum-wavevector space are continuous. The density of states in crystals can be derived from the dispersion relation between wave vector and energy [1]. Combining the free-electron approximation:

$$k = \frac{1}{\hbar} \sqrt{2mE} \quad (4)$$

yields the density-of-states function for free electrons [1, 2]:

$$g(E) = \frac{4\pi(2m)^{\frac{3}{2}}}{h^3} \sqrt{E} \quad (5)$$

Dispersion relations near conduction-band minima and valence-band maxima are approximately parabolic, corresponding to the second-order Taylor expansion of energy versus wave vector [1, 2]:

$$E = E_c + \frac{\hbar^2 k^2}{2m_n^*} \quad (6a)$$

$$E = E_v - \frac{\hbar^2 k^2}{2m_p^*} \quad (6b)$$

where k is the electron wave vector; m_n^* and m_p^* are the effective masses of electrons and holes at the conduction-band bottom and valence-band top, respectively. Using Equation (5), the density of states for electrons near the conduction-band minimum and holes near the valence-band maximum read [1, 2]:

$$g_c(E) = \frac{4\pi(2m_n^*)^{\frac{3}{2}}}{h^3} \sqrt{E - E_c} \quad (7a)$$

$$g_v(E) = \frac{4\pi(2m_p^*)^{\frac{3}{2}}}{h^3} \sqrt{E_v - E} \quad (7b)$$

E_c and E_v denote the conduction-band-minimum and valence-band-maximum energies. Combined with the Boltzmann approximation (Equations (2) and (3)), electron and hole concentrations are given by [1, 2]:

$$n = \int g_c(E) f(E) dE \quad (8a)$$

$$p = \int g_v(E) (1 - f(E)) dE \quad (8b)$$

After simplification [1, 2]:

$$n = N_c \exp\left(-\frac{E_c - E_f}{k_B T}\right) \quad (9a)$$

$$p = N_v \exp\left(-\frac{E_f - E_v}{k_B T}\right) \quad (9b)$$

where N_c and N_v are the effective density of states for the conduction band and valence band [1, 2]:

$$N_c = 2 \left(\sqrt{\frac{2\pi m_n^* k_B T}{h^2}} \right)^3 \quad (10a)$$

$$N_v = 2 \left(\sqrt{\frac{2\pi m_p^* k_B T}{h^2}} \right)^3 \quad (10b)$$

Equation (9) relates carrier concentrations to the energy difference between the Fermi level and band edges. These derivations ignore trap states originating from crystal defects and doping; thus carrier concentrations depend only on temperature. For a defect-free crystal under equilibrium, charge neutrality gives:

$$p = n = \sqrt{N_c N_v} = n_i \quad (11)$$

where n_i is defined as the intrinsic carrier concentration.

2.1.6 Quasi-Fermi Levels

Strictly speaking, the Fermi-level definition applies exclusively to thermal equilibrium. For analytical convenience, the formalism is often extended to non-equilibrium conditions, where E_f becomes quasi-Fermi levels. Electrons and holes generally possess distinct generation and injection channels under non-equilibrium and are therefore decoupled, requiring separate quasi-Fermi levels. From Equation (9), electron and hole quasi-Fermi levels read:

$$E_{fn} = E_c + k_B T \ln \frac{n}{N_c} \quad (12a)$$

$$E_{fp} = E_v - k_B T \ln \frac{p}{N_v} \quad (12b)$$

2.1.7 Mobility

Carriers drift directionally under electric fields. In steady-state conditions, drift velocity correlates with electric-field magnitude. Linear relations work for simple scenarios yet fail to reproduce phenomena such as current saturation in transistors. Especially under high electric fields, mobility degrades rapidly because of lattice thermal motion, defects, and carrier-carrier scattering. Two widely-adopted mobility models are presented below.

The Caughey-Thomas model describes saturation characteristics in transistors [3]:

$$\mu = \frac{\mu_{\text{eff}}}{\sqrt[\alpha]{1 + \left(\frac{\mu_{\text{eff}} \vec{E}_{\parallel}}{v_{\text{sat}}} \right)^\alpha}} \quad (13)$$

where μ_{eff} denotes low-field mobility; v_{sat} is carrier saturation velocity; $\vec{E}_{\parallel}$ is the component of the electric field parallel to the carrier-current direction; and α is the high-to-low-field transition coefficient.

The Arora model captures temperature-dependent mobility [4]:

$$\mu = \mu_0 \left(\frac{T}{T_c} \right)^{-\alpha_A} \quad (14)$$

in which μ_0 , T_c , and α_A are material-dependent fitting coefficients.

2.1.8 Einstein Relation

Besides drift driven by electric fields, carriers diffuse due to thermal agitation under concentration gradients. The diffusion coefficient D and mobility satisfy the Einstein relation:

$$\frac{D}{\mu} = \frac{k_B T}{e} \quad (15)$$

This equation demonstrates the tight coupling between drift and diffusion processes. Note that this expression is valid near band edges: states are sparsely filled in the conduction band and densely filled in the valence band. When quasi-Fermi levels penetrate deep into allowed bands, Equation (15) needs modification, which is addressed in Section 2.2.5.

2.1.9 Current Density

Once carrier concentrations, mobilities, and diffusion coefficients are known, current-density expressions for electrons and holes read:

$$\vec{J}_n = en\mu_n\vec{E} + eD_n\nabla n \quad (16a)$$

$$\vec{J}_p = ep\mu_p\vec{E} - eD_p\nabla p \quad (16b)$$

where μ_n , μ_p are electron and hole mobilities; D_n , D_p are electron and hole diffusion coefficients; and $\vec{E}$ is the electric-field vector. The first term in each formula represents drift current induced by the electric field; the second term corresponds to diffusion current originating from thermally-driven carrier equilibration.

2.2 Theoretical Analysis of Amorphous Semiconductors

Apart from long-range-ordered crystalline solids, polycrystalline materials (short-range order) and fully disordered amorphous solids can also exhibit semiconducting properties. Compared with crystals, the most prominent distinction for charge transport in amorphous systems is the reduced spatial delocalization of quantum states, manifested macroscopically as drastically lowered mobility.

Periodic-potential theory no longer applies for disordered structures. Physical models are therefore built upon quantum states of individual atoms, clusters, or molecules, and statistical-physical methods are employed to analyze electronic transitions among these quantum states. Concepts established for crystalline semiconductors (conductor, insulator, semiconductor, energy band, valence band, conduction band) are reused for amorphous materials with necessary revisions. For amorphous systems, the valence-band counterpart corresponds to Highest Occupied Molecular Orbitals (HOMOs), and the conduction-band counterpart corresponds to Lowest Unoccupied Molecular Orbitals (LUMOs). Whereas valence and conduction bands

consist of delocalized energy levels shared across lattice atoms, HOMOs and LUMOs represent sets of localized states belonging to molecular or cluster units available for electronic transitions.

Since microscopic particle arrangements in amorphous materials lack crystalline periodicity, carrier transport mechanisms differ substantially. Although current still decomposes into drift and diffusion contributions, other physical details must be treated separately.

Amorphous semiconductors are promising candidates for luminescent devices [5, 6]. Organic semiconductors, in particular, benefit from diverse molecular architectures and constitute excellent candidates for specialized sensors [7].

Consequently, results derived for crystalline semiconductors cannot be directly applied to amorphous systems; modifications are required to accommodate complex microstructures. Fundamental concepts for amorphous semiconductors, especially organic semiconductors, are introduced below.

2.2.1 Energy-State Structure

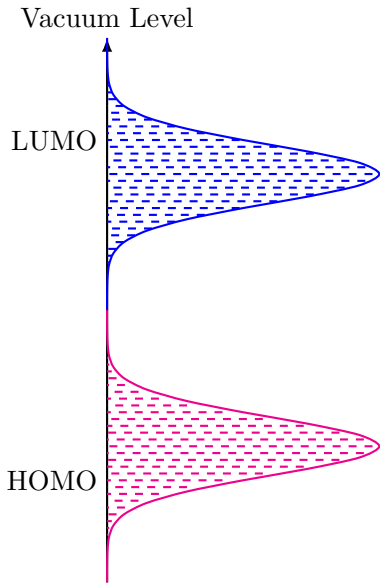


Figure 2: Schematic diagram of HOMO and LUMO state distributions

Because microscopic constituents in amorphous materials lack long-range crystalline order, Bloch-wave formalisms [1] are inapplicable, and fully delocalized crystal-wide electronic wave functions do not exist. Electrons predominantly occupy localized states associated with specific molecular or group moieties. Although identical moieties would possess nearly identical energy levels, exchange interactions, impurities, and defects disperse these energies around the isolated-molecule reference energies. Such state distributions centered around expectation values are commonly modeled using Gaussian statistics. Figure 2 illustrates HOMO-LUMO level distributions for organic semiconductors. The corresponding Gaussian density-of-states (DOS)

functions read:

$$g_L(E_L, E_{Lc}, \sigma_L) = \frac{N_L}{\sigma_L \sqrt{2\pi}} \exp\left(-\frac{(E_L - E_{Lc})^2}{2\sigma_L^2}\right) \quad (17a)$$

$$g_H(E_H, E_{Hc}, \sigma_H) = \frac{N_H}{\sigma_H \sqrt{2\pi}} \exp\left(-\frac{(E_H - E_{Hc})^2}{2\sigma_H^2}\right) \quad (17b)$$

Here E_L and E_H denote individual energy levels within LUMOs and HOMOs; N_L , N_H are total densities of states; σ_L , σ_H are distribution widths (variances squared); E_{Lc} , E_{Hc} are distribution central energies (expectation values). HOMO and LUMO states appear in pairs such that $N_L = N_H$, whose magnitude is determined by molecular or polymer segment counts, typically ranging from $10^{20} \sim 10^{21} \text{ cm}^{-3}$ [8]. Typical values for σ_L and σ_H lie within $0.05 \sim 0.15 \text{ eV}$ [9, 10]. The energy difference $E_{Lc} - E_{Hc}$ defines the material band gap, around 1.0 eV for solar-cell absorbers [11], and can exceed 10 eV for insulating polymers [12].

2.2.2 Carrier Concentration

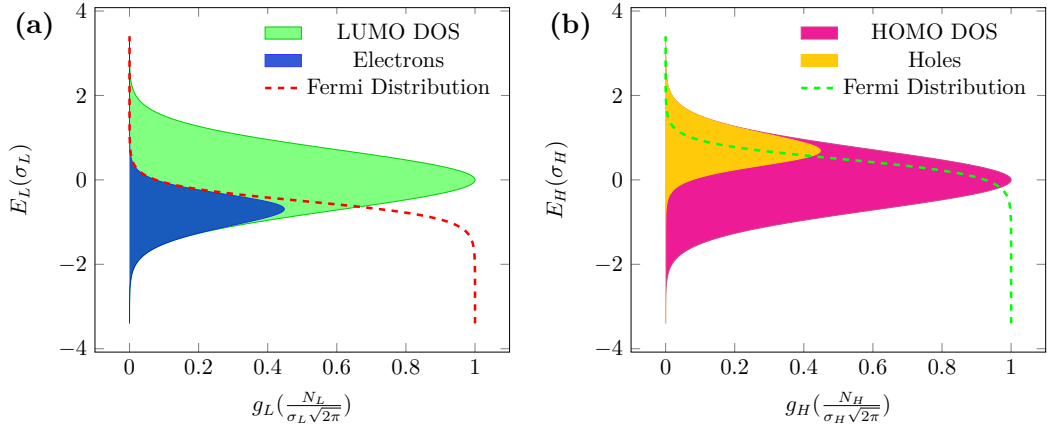


Figure 3: Schematic diagrams for electrons (a) and holes (b) within Gaussian LUMO-HOMO DOS

Boltzmann approximations are convenient for crystalline semiconductors. For amorphous semiconductors without well-defined band edges, the Boltzmann approximation breaks down, and the full Fermi-Dirac distribution must be adopted, as shown in Figure 3. Analogous to Equation (8), electron and hole concentrations for amorphous semiconductors read:

$$n = \int_{-\infty}^{\infty} g_L(E_L, E_{Lc}, \sigma_L) f(E_L, E_f) dE_L \quad (18a)$$

$$p = \int_{-\infty}^{\infty} g_H(E_H, E_{Hc}, \sigma_H) (1 - f(E_H, E_f)) dE_H \quad (18b)$$

These integrals multiply Gaussian DOS functions with Fermi-Dirac distributions, known as Gaussian-Fermi integrals [13, 14]. No closed-form analytical solution exists; solutions rely on numerical evaluation [15] or approximate expressions [14]. The resulting behavior is plotted in

Figure 4. At low carrier densities, carrier concentrations vary exponentially with Fermi-level offset similar to crystalline semiconductors. At high densities, saturation emerges, consistent with the relatively low total density of states typical of organic amorphous semiconductors. For later reference, we define compact notations for the two integrals in Equation (18):

$$n = N_L \text{Gfi}_L(E_f, E_{Lc}, \sigma_L) \quad (19a)$$

$$p = N_H \text{Gfi}_H(E_f, E_{Hc}, \sigma_H) \quad (19b)$$

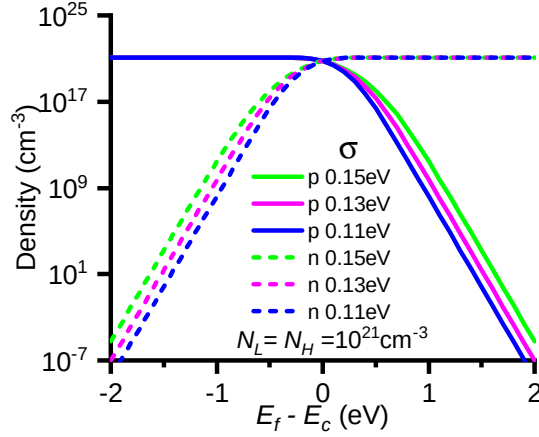


Figure 4: Electron and hole concentrations versus energy offset between Fermi level and Gaussian DOS central energies

2.2.3 Quasi-Fermi Levels

Identical to crystalline semiconductors, electrons and holes in amorphous semiconductors can possess separate generation, injection, and recombination pathways, so their concentrations evolve asynchronously. Consequently, multiple quasi-Fermi levels are required to describe non-equilibrium states. Define iGfi_L and iGfi_H as inverse functions of the Gaussian-Fermi integrals in Equation (19). The quasi-Fermi levels are expressed as:

$$E_{fn} = \text{iGfi}_L\left(\frac{n}{N_L}, E_{Lc}, \sigma_L\right) \quad (20a)$$

$$E_{fp} = \text{iGfi}_H\left(\frac{p}{N_H}, E_{Hc}, \sigma_H\right) \quad (20b)$$

2.2.4 Mobility

Stronger electron localization in amorphous semiconductors suppresses band-like transport; charge transport is dominated by hopping transitions between distinct localized states, as illustrated in Figure 5. Several widely-used mobility models for amorphous semiconductors are summarized below.

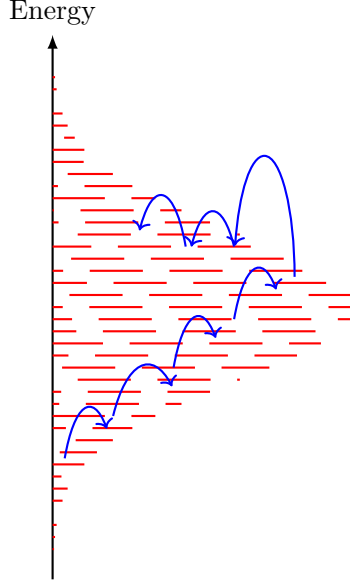


Figure 5: Schematic diagram of charge transport among amorphous localized states

Poole–Frenkel model [16, 17]:

$$\mu = \mu_0 \exp \left[-\frac{\Delta}{k_B T} + \left(\frac{\beta}{k_B T} - \gamma \right) \sqrt{|\vec{E}|} \right] \quad (21)$$

where μ_0 is the pre-factor mobility; β and γ are fitting parameters; Δ denotes the effective activation energy. This model originates from the Poole-Frenkel effect [18]: applied electric fields lower the effective hopping barrier and enhance directed charge motion.

Micro-conductivity model [19, 20]:

$$\mu_p = \frac{\Gamma_p}{ep} \exp \left(\frac{E_f}{k_B T} \right) \quad (22)$$

where Γ_p is the conductivity prefactor. Built upon the variable-range-hopping picture [21], this model captures how the Fermi-level position modulates carrier density and hopping rates: high density-of-states regions correspond to low hopping barriers and higher mobility, and vice-versa.

Carrier-concentration-dependent model [9, 22]

$$\mu = v_0 \frac{e}{k_B T} \frac{3B_c}{4\pi R(\varepsilon_t)p} \exp \left[-\frac{2B_c^{1/3}}{a} R(\varepsilon_t) - \frac{\varepsilon_t - E_F}{k_B T} \right] \quad (23)$$

where ε_t is the transition energy; B_c is a fitting coefficient; $R(\varepsilon_t)$ denotes the average hopping distance. This model describes charge hopping within a finite energy window ε_t ; transitions for states below this threshold require multi-step excitation and exhibit low transition probabilities [9].

2.2.5 Generalized Einstein Relation

Derivation of the standard Einstein relation Equation (15) assumes charge-neutrality conditions for crystalline semiconductors [2]. In amorphous semiconductors, larger intermolecular

distances reduce the total density-of-states significantly. Electrostatic potentials from ionized sites left behind after carrier generation must be incorporated, leading to the generalized Einstein relation.

Taking holes as an example: under equilibrium conditions, drift and diffusion currents cancel such that net carrier concentration remains constant. From Equation (16b):

$$\begin{aligned} 0 &= p\mu_p\vec{E} - D_p\nabla p = -p\mu_p\nabla\varphi - D_p\nabla p \\ &= (-p\mu_p - D_p\frac{dp}{d\varphi})\nabla\varphi \end{aligned} \quad (24)$$

where φ is the local electrostatic potential. The Fermi level shifts with potential: $dE_f = -ed\varphi$. The negative sign reflects the definition of Fermi level for negatively-charged electrons. Substituting this relation yields [10]:

$$D_p = \frac{\mu_p p}{e} \frac{1}{\frac{dp}{dE_f}} \quad (25)$$

Similarly for electrons:

$$D_n = \frac{\mu_n n}{e} \frac{1}{\frac{dn}{dE_f}} \quad (26)$$

These expressions constitute the generalized Einstein relations for electrons and holes. Diffusion coefficients depend strongly on carrier concentration. Especially for low-density-of-states amorphous materials such as organic semiconductors, diffusion coefficients rise sharply at high carrier densities and broaden space-charge regions.

2.3 Analytical Methods for Defect States in Organic Semiconductors

2.3.1 Gaussian and Exponential Distributions

For both organic and inorganic semiconductors, trap-state energy distributions are commonly described by Gaussian functions with characteristic energy width (Equation (27)) or exponential functions with characteristic temperature (Equation (28)) [23–27].

Gaussian distribution:

$$g = \frac{N_T}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(E - E_c)^2}{2\sigma^2}\right) \quad (27)$$

where N_T is total trap density; σ is energy standard deviation; E_c is the central trap energy.

Exponential distribution:

$$g = N_{t_0} \exp\left(\frac{E - E_t}{k_B T_t}\right) \quad (28)$$

where N_{t_0} denotes effective trap density; k_B is Boltzmann constant; T_t represents the characteristic temperature for trap distribution.

Owing to simpler mathematical treatment, exponential distributions are widely adopted in traditional semiconductor analysis. Nevertheless, they lack saturation behavior and are unsuitable for trap analysis under high-bias conditions.

In organic semiconductors, charge carriers behave differently compared with crystalline inorganic semiconductors. Instead of atoms in silicon lattices, organic systems consist of delocalized molecular moieties with larger intermolecular separations and weaker wave-function

overlap. Charge transport proceeds via transitions between molecular energy levels rather than lattice-delocalized band transport, leading to much lower mobilities. Instead of sharply-defined conduction and valence bands [28], organic semiconductors possess distributed energy manifolds: LUMOs for electron transport and HOMOs for hole transport. Gaussian distributions are the preferred description for these density-of-states manifolds [29].

Because sharp band edges disappear in organic semiconductors, the Boltzmann approximation [2] becomes inadequate for describing electron occupation; the full Fermi–Dirac statistics must be used [9]. Differences in density-of-states and occupation statistics between organic and inorganic semiconductors demand modified treatments for trap generation and recombination. Below we analyze generation-recombination processes for donor-like traps in organic semiconductors, using Miller-Abrahams hopping rates together with Fermi-Dirac statistics.

2.3.2 Distribution Functions and Occupation of Donor-Like Trap States

Trap states within the band gap of organic semiconductors are classified as donor-like or acceptor-like [30]. Their statistical distributions are modeled with Gaussian functions. The density-of-states for donor-like traps reads:

$$g_D(E_D, E_{Dc}, \sigma_D) = \frac{N_D}{\sigma_D \sqrt{2\pi}} \exp\left(-\frac{(E_D - E_{Dc})^2}{2\sigma_D^2}\right) \quad (29)$$

where E_D denotes trap energy; N_D , σ_D , E_{Dc} are total trap density, energy-distribution width (standard deviation), and central trap energy, respectively. In subsequent text, $g_D(E_D)$ serves as shorthand for $g_D(E_D, E_{Dc}, \sigma_D)$.

Neutral donor-like traps contain an occupied electron; upon electron emission, positively-charged empty trap states remain. The density of positively-charged empty donor-like traps is determined by unoccupied trap states:

$$p_D = \int_{-\infty}^{\infty} g_D(E_D, E_{Dc}, \sigma_D)(1 - f(E_D, E_{Df}))dE_D \quad (30)$$

2.3.3 Miller-Abrahams Formula

Two major theoretical frameworks describe charge hopping in organic materials: Marcus transition-rate theory [31,32] and the Miller-Abrahams rate model [33]. Marcus theory provides rigorous physical foundations for polaron generation and annihilation yet involves numerous complicated parameters. The Miller-Abrahams formula, a variant of the Arrhenius law [34], is more widely applied. It has been extensively employed to investigate charge-transport mobilities [8, 20, 35–37], trap generation-recombination [38, 39], magnetoresistance from spin-flip processes [40–42], bipolaron transport [43], and multi-phonon excitation dynamics [34]. The Miller-Abrahams formula gives the transition probability for an electron hopping from an initial state with energy E_i to an unoccupied final state with energy E_j [33, 38]:

$$v_{ij} = v_0 \exp\left(-2\frac{R_{ij}}{a_i} - \frac{E_j - E_i + |E_j - E_i|}{2k_B T}\right) \quad (31)$$

where v_0 is the attempt-to-hop frequency; a_i is the localization radius of the initial state; R_{ij} is the spatial separation between initial and final sites.

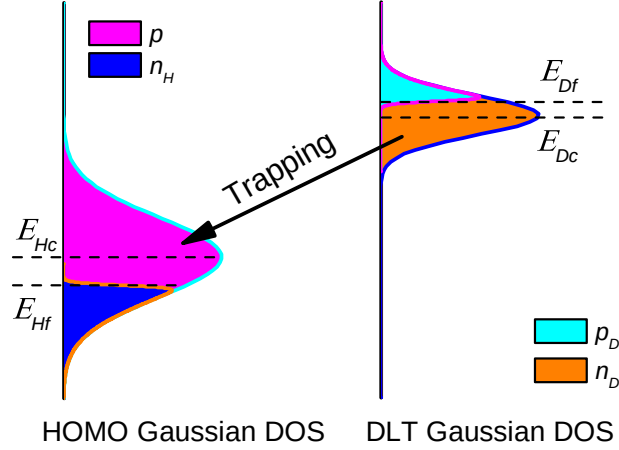


Figure 6: Schematic diagram for trap-charge generation of donor-like traps

2.3.4 Trap-Charge Generation Process

During trap-charge generation for donor-like traps in organic semiconductors, electrons hop from occupied donor-like trap states onto unoccupied HOMO sites, as sketched in Figure 6. Donor-like traps are deep-level states [30]; here we assume their energies lie above HOMO levels.

Integrating over all occupied donor-like-trap electrons (orange region in Figure 6) and empty HOMO states (purple region in Figure 6) weighted by Miller-Abrahams hopping rates yields the total trap-charge generation rate:

$$k_T = \iint_{-\infty}^{\infty} v_{DH} g_D(E_D) f(E_D, E_{Df}) g_H(E_H) (1 - f(E_H, E_{Hf})) dE_D dE_H \quad (32)$$

where E_D and E_H are energies of donor-like-trap and HOMO states, respectively; v_{DH} is the hopping rate for electrons from E_D to E_H :

$$v_{DH} = v_0 \exp\left(-2 \frac{R_{DH}}{\alpha_D}\right) \quad (33)$$

α_D denotes the localization length for donor-like trap states; R_{DH} is the average spatial separation between donor-like traps and HOMO sites.

Since integration variables E_D and E_H are independent in Equation (32), the double integral factors and simplifies to:

$$k_T = v_{DH} n_D p \quad (34)$$

where p is the hole concentration within HOMOs (Equation (??)); n_D is the total density of electron-occupied donor-like trap states (orange region in Figure 6):

$$\begin{aligned} n_D &= \int_{-\infty}^{\infty} g_D(E_D, E_{Dc}, \sigma_D) f(E_D, E_{Df}) dE_D \\ &= N_D - p_D \end{aligned} \quad (35)$$

N_D is the total density of donor-like traps; p_D is the density of empty trap states defined in Equation (30).

2.3.5 Trap-Charge Recombination Process

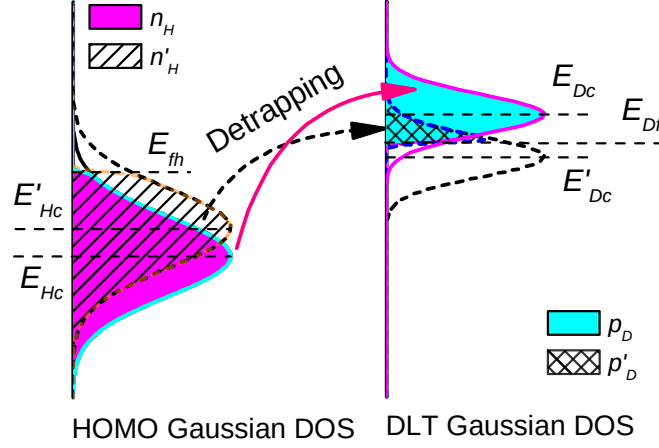


Figure 7: Schematic diagram for trap-charge recombination of donor-like traps

Trap-charge recombination corresponds to positively-charged empty donor-like traps recapturing electrons. Although LUMO energies lie above donor-like trap levels, the Fermi level is far below LUMOs under hole-accumulation conditions, and LUMO states remain unpopulated. Recombination electrons originate from HOMO states (red arrow in Figure 7). The recombination rate is computed by integrating over occupied HOMO electron densities (purple region in Figure 7), empty donor-like-trap densities (cyan region in Figure 7), and Miller-Abrahams hopping rates:

$$k_D = \iint_{-\infty}^{\infty} v_{HD} g_H(E_H) f(E_H, E_{Hf}) g_D(E_D) (1 - f(E_D, E_{Df})) dE_H dE_D \quad (36)$$

where v_{HD} is the Miller-Abrahams hopping rate for electrons from HOMO level E_H onto donor-like trap level E_D :

$$v_{HD} = v_0 \exp\left(-2\frac{R_{DH}}{a_H} - \frac{E_D - E_H}{k_B T}\right) \quad (37)$$

a_H denotes the electron localization length inside HOMO states.

Define:

$$v_{HD0} = v_0 \exp\left(-2\frac{R_{DH}}{a_H}\right) \quad (38)$$

Equation (36) becomes:

$$k_D = \iint_{-\infty}^{\infty} v_{HD0} \exp\left(-\frac{E_D - E_H}{k_B T}\right) g_H(E_H) f(E_H, E_{Hf}) g_D(E_D) (1 - f(E_D, E_{Df})) dE_H dE_D \quad (39)$$

2.3.6 Effective Unoccupied and Occupied Densities

The extra exponential factor $\exp\left(-\frac{E_D - E_H}{k_B T}\right)$ in Equation (39) permits factorization:

$$k_D = v_{HD0} \int_{-\infty}^{\infty} \exp\left(\frac{E_H}{k_B T}\right) g_H(E_H) f(E_H, E_{Hf}) dE_H \int_{-\infty}^{\infty} \exp\left(-\frac{E_D}{k_B T}\right) g_D(E_D) (1 - f(E_D, E_{Df})) dE_D \quad (40)$$

Evaluate the HOMO-related integral by substituting the Gaussian density-of-states:

$$\begin{aligned} & \int_{-\infty}^{\infty} \exp\left(\frac{E_H}{k_B T}\right) g_H(E_H) f(E_H, E_{Hf}) dE_H \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{E_H}{k_B T}\right) \frac{N_H}{\sigma_H \sqrt{2\pi}} \exp\left(-\frac{(E_H - E_{Hc})^2}{2\sigma_H^2}\right) f(E_H, E_{Hf}) dE_H \\ &= \int_{-\infty}^{\infty} \frac{N_H}{\sigma_H \sqrt{2\pi}} \exp\left(-\frac{(E_H - E_{Hc})^2}{2\sigma_H^2} + \frac{E_H}{k_B T}\right) f(E_H, E_{Hf}) dE_H \\ &= \exp\left(\frac{\sigma_H^2}{2k_B T^2} + \frac{E_{Hc}}{k_B T}\right) \int_{-\infty}^{\infty} g_H(E_H, E_{Hc} + \frac{2}{k_B T}, \sigma_H) f(E_H, E_{Hf}) dE_H \\ &= \exp\left(\frac{\sigma_H^2}{2k_B T^2} + \frac{E_{Hc}}{k_B T}\right) n'_H \end{aligned} \quad (41)$$

where:

$$n'_H = \int_{-\infty}^{\infty} g_H(E_H, E'_{Hc}, \sigma_H) f(E_H, E_{Hf}) dE_H \quad (42)$$

and

$$E'_{Hc} = E_{Hc} + \frac{\sigma_H^2}{k_B T} \quad (43)$$

n'_H shares the mathematical form of the electron-density expression for LUMO states, Equation (??). We term n'_H the effective occupied density for HOMO states, and E'_{Hc} the effective central HOMO energy. The purple region and dashed line in Figure 7 illustrate the energy shift from the original HOMO central energy to the effective value.

For the donor-like-trap integral in Equation (40):

$$\begin{aligned} & \int_{-\infty}^{\infty} \exp\left(-\frac{E_D}{k_B T}\right) g_D(E_D) (1 - f(E_D, E_{Df})) dE_D \\ &= \int_{-\infty}^{\infty} \exp\left(-\frac{E_D}{k_B T}\right) \frac{N_D}{\sigma_D \sqrt{2\pi}} \exp\left(-\frac{(E_D - E_{Dc})^2}{2\sigma_D^2}\right) (1 - f(E_D, E_{Df})) dE_D \\ &= \exp\left(\frac{\sigma_D^2}{2k_B T^2} - \frac{E_{Dc}}{k_B T}\right) \int_{-\infty}^{\infty} g_D(E_D, E_{Dc} - \frac{2}{k_B T}, \sigma_D) (1 - f(E_D, E_{Df})) dE_D \\ &= \exp\left(\frac{\sigma_D^2}{2k_B T^2} - \frac{E_{Dc}}{k_B T}\right) p'_D \end{aligned} \quad (44)$$

where:

$$p'_D = \int_{-\infty}^{\infty} g_D(E_D, E'_{Dc}, \sigma_D) (1 - f(E_D, E_{Df})) dE_D \quad (45)$$

and

$$E'_{Dc} = E_{Dc} - \frac{\sigma_D^2}{k_B T} \quad (46)$$

p'_D is defined as the effective empty-state density for donor-like traps; E'_{Dc} denotes the effective central trap energy. The cyan region and dashed line in Figure 7 visualize this energy shift.

Combining results from Equation (41) and Equation (44), Equation (40) transforms into:

$$k_D = c_E v_{DH0} \iint_{-\infty}^{\infty} g_H(E_H, E'_{Hc}, \sigma_H) f(E_H, E_{Hf}) g_D(E_D, E'_{Dc}, \sigma_D) (1 - f(E_D, E_{Df})) dE_H dE_D \quad (47)$$

with prefactor c_E :

$$c_E = \exp\left(\frac{\sigma_H^2 + \sigma_D^2}{2k_B^2 T^2} + \frac{E_{Hc} - E_{Dc}}{k_B T}\right) \quad (48)$$

Analogous to Equation (32), Equation (47) reduces to a product-form expression similar to Equation (34):

$$k_D = c_E v_{DH0} n'_H p'_D \quad (49)$$

To summarize: starting from Fermi-Dirac statistics and the Miller-Abrahams hopping formula, we obtain analytical expressions for trap-charge generation and recombination rates for donor-like traps. By introducing effective occupied and unoccupied densities for HOMO and donor-like-trap manifolds, the complicated double-integral recombination rate simplifies to a compact product expression, Equation (49). As sketched by black dashed arrows in Figure 7, the hopping-barrier effect is recast as energy-center shifts for HOMO and donor-like traps, while quasi-Fermi-level positions remain unchanged. The HOMO manifold shifts upward by $\frac{\sigma_H^2}{k_B T}$ (Equation (43)), and donor-like-trap manifold shifts downward by $\frac{\sigma_D^2}{k_B T}$ (Equation (46)). Occupied-state and empty-state densities evaluated at these shifted effective energies reproduce the hopping-rate integral result. This mathematical simplification is convenient for TCAD simulation workflows, since TCAD solves for particle densities directly. Subsequent sections discuss coupling this formulation into TCAD and simulating organic thin-film field-effect transistors based on polymer semiconductor DPP-DTT.

2.3.7 Net Generation Rate for Trap Charges

To compute the time evolution of donor-like trap-charge density:

$$\frac{\partial p_D}{\partial t} = k_T - k_D \quad (50)$$

Three charge components enter the Poisson equation (Equation (84)): electrons, holes, and ionized donor-like trap charges.

Organic field-effect transistors typically turn on under hole accumulation and turn off under depletion conditions. We only consider electron hopping between donor-like traps and HOMO states and neglect transitions involving LUMOs. The net hole generation rate is:

$$G_p = -(k_T - k_D) \quad (51)$$

2.3.8 Conversion between Areal and Volume Trap Densities

Field-effect-transistor characterization commonly extracts trap information from subthreshold swing and threshold-voltage V_{th} measurements [44, 45]. These experimental observables are

interpreted in terms of areal trap densities, whereas our formulation employs volume trap densities. The conversion formula reads:

$$N_D = \frac{N_{Ds}}{d_D} \exp\left(-\frac{x}{d_D}\right) \quad (52)$$

where x is depth within the semiconductor layer; d_D is the characteristic thickness describing the spatial profile of donor-like traps. Integrating over depth recovers the total areal trap density N_{Ds} .

2.4 Basic Theory of Ferroelectric Materials

Ferroelectric materials are defined by the existence of spontaneous electric polarization. This definition imposes no restrictions on electrical conductivity; reported ferroelectrics can be conductors [46], semiconductors [47], or insulators. Most practical ferroelectric applications exploit remnant polarization and focus on insulating ferroelectric materials. Accordingly, the following theoretical treatment assumes insulating ferroelectrics.

The electric displacement field inside an insulator decomposes into two contributions:

$$\vec{D} = \varepsilon_0 \vec{E} + \vec{P} \quad (53)$$

For ordinary insulators, polarization $\vec{P}$ depends linearly on the electric field $\vec{E}$. Certain materials exhibit non-linear behavior together with remnant polarization: finite polarization $\vec{P} \neq 0$ persists even when $\vec{E} = 0$.

Within TCAD simulation, two primary theoretical frameworks relate polarization and electric field for ferroelectrics: the Landau-Ginsberg phase-transition theory and empirical formulas proposed by Miller et al. [48]. Landau-Ginsberg theory analyzes phase-transition dynamics from free-energy-polarization relations and supports time-dependent simulations at the cost of higher computational complexity. Miller's empirical model fits experimental measurements accurately with comparatively low implementation overhead and constitutes the mainstream approach for TCAD. Both frameworks are briefly introduced below.

2.4.1 Landau-Ginsberg Theory

According to Landau-Ginsberg ferroelectric phase-transition theory [49, 50], under isothermal and mechanically-free conditions and considering only one polarization direction, the free-energy density G_i of ferroelectric and paraelectric phases expands in powers of polarization P_i :

$$G_i = -E_i P_i + G_{i0} + \frac{1}{2} \alpha (T - T_0) P_i^2 + \frac{1}{4} \beta P_i^4 + \frac{1}{6} \gamma P_i^6 + \quad (54)$$

$(i = 1, 2, 3)$

where G_{i0} is the free-energy offset at $P_i = 0$; T_0 denotes the Curie-Weiss temperature; α , β , γ are material constants.

$\beta > 0$ corresponds to second-order ferro-para phase transitions; $\beta < 0$ describes first-order transitions accompanied by latent heat and multiple characteristic temperatures. We focus on second-order phase transitions, whose free-energy-polarization relations appear in Figure 8(a).

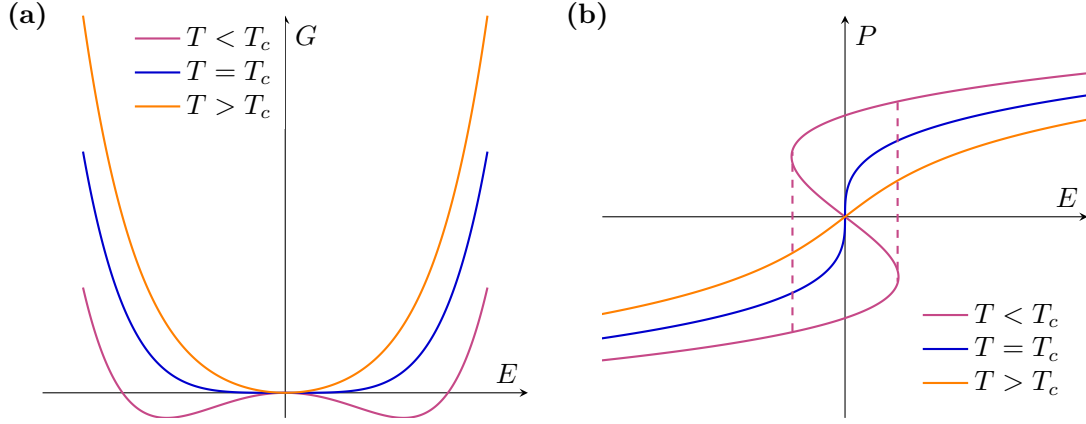


Figure 8: Landau-Ginsberg theoretical curves for ferroelectrics under different temperatures: (a) free-energy versus polarization; (b) polarization versus electric-field characteristics

For temperatures below Curie temperature T_c , two free-energy minima correspond to two spontaneous-polarization states (ferroelectric phase). When $T \geq T_c$, only one minimum survives and remnant polarization vanishes (paraelectric phase).

Minimizing free-energy gives the equilibrium relation between electric field and polarization:

$$\frac{\partial G}{\partial P} = 0 = -E + \alpha(T - T_0)P_i + \beta P_i^3 + \gamma P_i^5 \quad (55)$$

Figure 8(b) plots polarization-electric-field curves across the phase-transition temperature. For $T < T_0$, the $E(P)$ curve possesses one maximum and one minimum, generating the hysteresis loop shown by the magenta curve in Figure 8(b).

Time-dependent evolution of spontaneous polarization follows the time-dependent Landau-Ginsberg equation [51, 52]:

$$\frac{dP_i}{dt} = -L \frac{\partial G}{\partial P_i} \quad (56)$$

where coefficient L relates to domain-wall-motion kinetics and thermodynamic properties. Figure 9 plots polarization-voltage curves for ferroelectric thin films computed within the Landau-Ginsberg framework.

2.4.2 Miller Empirical Formula

Although time-dependent Landau-Ginsberg equations capture ferroelectric thermodynamic mechanisms, their output often deviates from experimentally-measured hysteresis loops. Miller's empirical formula is widely-adopted to reproduce experimental hysteresis characteristics with simpler computational overhead [20, 48, 53]. In this model, polarization decomposes along coordinate axes:

$$P_i = P_s c_i \tanh[\omega(E_i - E_{FC})] + P_s c_{\text{off}_i} \quad (57)$$

where $i \in (x, y, z)$; $c_{\text{off}_i} \in (-1, 1)$ and $c_i \in (0, 1]$ are history-dependent coefficients; and

$$\omega = \ln \frac{(P_s + P_r)}{(P_s - P_r)} / (2E_{FC}), \quad (58)$$

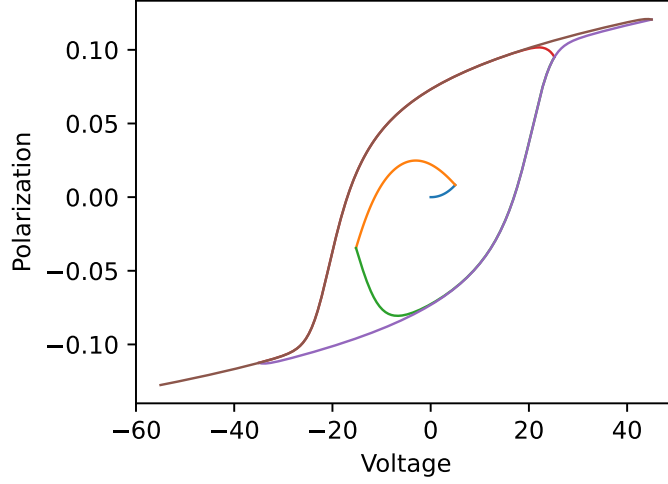


Figure 9: Polarization-voltage curve under Landau-Ginsberg model

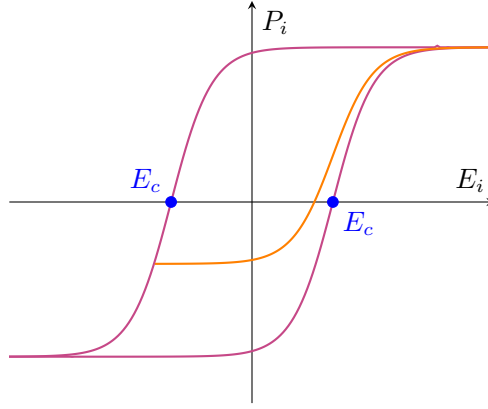


Figure 10: Ferroelectric hysteresis loop simulated by Miller's empirical formula

P_s , P_r , E_{FC} denote saturation polarization, remnant polarization, and coercive field, respectively.

The relation between c_i and c_{off_i} reads:

$$c_i \cdot \text{sign}(E_{FC}) + c_{\text{off}_i} = \text{sign}(E_{FC}) \quad (59)$$

where the $\text{sign}()$ function returns +1 for rising electric fields and -1 for falling electric fields. Simulation results are illustrated in Figure 10.

Besides Miller's empirical formulation, differential permittivity approaches appear in literature and commercial simulation software to describe polarization-electric-field relations [54–56]. Differential permittivity is defined as:

$$\varepsilon_{\text{diff}_i} = \frac{dP_i}{dE_i} \quad (60)$$

While convenient for many scenarios, differential permittivity introduces mathematical singularities when reversing electric-field direction. For this reason, Miller's empirical formula is more frequently employed.

3 Numerical Simulation Methods for Semiconductor-Ferroelectric Devices

Numerical simulation methods analyze the spatial distribution of physical parameters based on partial-differential equations describing device-level physical processes. By comparing computed macroscopic observables such as current, capacitance and voltage against real-world device measurements, simulation guides the development of semiconductor manufacturing processes. Moreover, numerical simulation offers unique advantages for investigating novel-structure devices. For newly-proposed semiconductor devices, one only needs to define the device geometry and material distribution, and simulation can compute both microscopic physical-quantity distributions and macroscopic electrical output characteristics. This capability greatly reduces the time, technical barriers and experimental costs required to validate novel device concepts. It also supplies reliable raw reference data for traditional analytical approximate formulas and improves researchers' theoretical analysis efficiency.

3.1 Core Concepts of Semiconductor Device Simulation

Even with well-established theoretical frameworks, performing numerical analysis for practical physical problems remains non-trivial. First, physical processes are described by diverse theorems and formulas; selecting the appropriate physical relations for solving a given problem is itself a non-trivial task. From a more general perspective, nearly all physical processes obey conservation laws: for example, conservation of mass, momentum, charge, and amount-of-substance.

These conservation laws can be expressed via the following continuity-equation form:

$$\iiint_V \frac{\partial X}{\partial t} dV = - \oint_S \vec{A} \cdot d\vec{S} + \iiint_V G dV \quad (61)$$

where V denotes the finite volume of interest for the conservation problem; X is the spatial distribution function of the conserved quantity; t represents time; $\vec{S}$ denotes the outer surface of the domain with outward-pointing normal vectors; $\vec{A}$ is the areal flux rate describing the flow of conserved quantities across the domain boundary; G is the total volumetric generation-recombination rate of the conserved quantity within the volume.

The corresponding differential-form equation reads:

$$\frac{\partial X}{\partial t} = -\nabla \cdot \vec{A} + G \quad (62)$$

where $\nabla \cdot$ is the divergence operator for vector fields.

There exist conceptual similarities between semiconductor carrier transport and hydrodynamics, including carrier flux and Gauss' s law for electric fields. Accordingly, numerical semiconductor simulation can borrow analysis techniques from Computational Fluid Dynamics (CFD).

Common spatial-discretization approaches include the finite-volume method, finite-element method, finite-difference method, and boundary-element method. Among these, the **finite-volume method** works directly from the continuity equation for conserved physical quantities (Equation (61)) and carries clear physical meaning. It can also be combined with the Scharfetter-Gummel discretization scheme (3.4.3) to accurately resolve semiconductor features featuring

steep field gradients. For these reasons it is widely adopted and has become the dominant discretization technique in TCAD device simulation.

3.1.1 Mesh Discretization

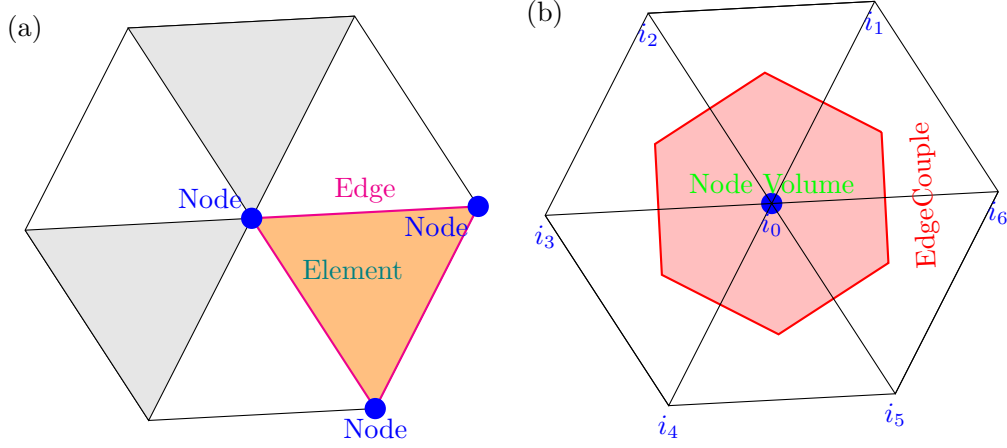


Figure 11: Schematic of meshes

To clarify the principles behind the finite-volume method, we introduce fundamental mesh-related definitions for two-dimensional problems [57]¹:

- a. **Node**: Discrete point representative of physical quantities over its local spatial neighborhood when continuous domains are discretized, see Figure 11(a).
- b. **Edge**: Line segment connecting two adjacent nodes of a geometric element. Physical quantities with differential character (electric field, current density, flow velocity, diffusion gradient, etc.) are evaluated on edges, see Figure 11(a).
- c. **Triangle Element**: Simple geometric primitive bounded by neighboring nodes and containing no interior nodes; examples include 2-D triangular elements (orange region in Figure 11(a)) and 3-D tetrahedral elements.
- d. **Edge Couple**: Segments formed by perpendicular bisectors of element edges intersecting one another, as illustrated by red segments in Figure 11(b).
- e. **Couple Element**: Closed region bounded by Edge-Couple segments originating from all edges connected to a node; this is the actual integration domain for finite-volume computations, shown as the pink region in Figure 11(b).
- f. **Node Volume**: Spatial volume of a Couple Element.
- g. **Mesh**: Network of geometric elements sharing nodes, representing the computational geometry. Different domain-splitting strategies yield distinct meshes.
- h. **Duality (dual partition)**: Re-partitioning of the original mesh using Couple-Element domains, also known as circumcenter-based partitioning; see pink region in Figure 11(b).

¹Some terms have no established standardized Chinese terminology and are translated literally from English.

- i. **Degree of Freedom:** Number of unknown variables defined at each node, e.g. electrostatic potential, carrier concentrations, temperature in semiconductor simulation.

3.1.2 Finite-Volume Discretization

The core idea of the finite-volume method is to discretize continuity equations. For the control volume indexed i in Figure 11(b), Equation (61) becomes:

$$\frac{\partial X_i}{\partial t} \cdot V_i = - \sum_{k=1}^6 \vec{A}_{i_0 i_k} \cdot \vec{L}_{i_0 i_k} + G_i \cdot V_i \quad (63)$$

where X_i is the quantity density at node i , V_i denotes node volume, $\vec{A}_{i_0 i_k}$ is the flux-density vector from node i toward node i_k , $\vec{L}_{i_0 i_k}$ is the area vector of the Edge-Couple face oriented along the outward normal, and G_i is the net volumetric generation rate at node i . This equation assumes: 1). Uniform density and generation rate within each control volume, represented by X_i and G_i ; 2). Uniform flux density across each Edge-Couple face, represented by $A_{i j_k}$.

After obtaining the discretized relation (63), one builds a system of algebraic equations whose unknowns are nodal-domain variables X_i ²:

$$\mathcal{F}(\mathcal{X}) = \mathcal{R} \quad (64)$$

Here $\mathcal{X}$ is the solution-space vector, $\mathcal{F}(\mathcal{X})$ represents the assembled residual system, and $\mathcal{R}$ is the right-hand-side vector. The target solution $\mathcal{X}$ satisfies:

$$\mathcal{F}(\mathcal{X}) = 0 \quad (65)$$

Node counts in practical simulations often exceed 10^5 . Direct matrix-solving becomes infeasible; iterative solvers are preferred. The Newton-Raphson method is widely used [58], together with Jacobi iteration and Gauss-Seidel iteration [59]. Assembly of algebraic systems and details of Newton-Raphson solving are elaborated in Chapter 3, Section 3.3.2.

3.2 Conventional Flux Evaluation in Finite-Volume Method

From the finite-volume discretization (63), the total outward flux across the control-volume boundary for node i_0 in Figure 12 reads:

$$F_T = \sum_{k=1}^5 \vec{A}_{i_0 i_k} \cdot \vec{L}_{i_0 i_k} \quad (66)$$

Here $\vec{L}_{i_0 i_1}$ denotes the Edge-Couple vector between node i_0 and i_1 , oriented along the outward normal of the control volume at node i_0 . The magnitude (area in three-dimensional problems) of $\vec{L}_{i_0 i_1}$ is determined by mesh geometry. Only the flux-density vector $\vec{A}_{i_0 i_k}$ needs to be computed to obtain total flux.

Although the assumption of uniform flux density over Edge-Couple faces (Section 3.1.2) simplifies calculations, evaluating flux vectors on these faces remains non-trivial. For flux originating from concentration gradients, the simplified approach illustrated in Figure 12 is widely adopted.

²Bold calligraphic uppercase symbols denote functions or vectors in solution space.

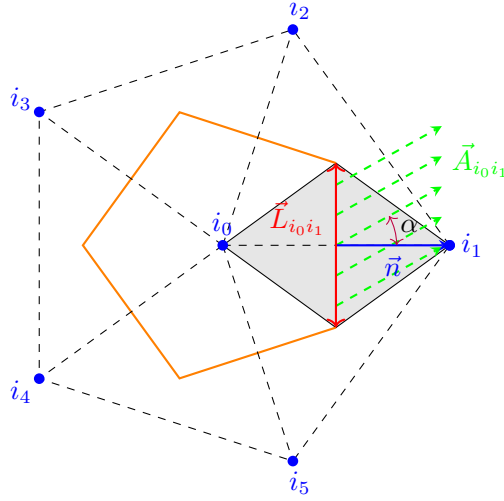


Figure 12: Diagram for traditional method for flux evaluation in FVM.

For nodes i_0 and i_1 , let scalar-field values at these nodes be X_0 and X_1 , respectively. Let $\vec{A}_{i_0i_1}$ be the true flux density across Edge-Couple $L_{i_0i_1}$. The discrete relation becomes:

$$\begin{aligned} \vec{A}_{i_0i_1} \cdot \vec{n}_{i_0i_1} &= \cos \alpha \left| \vec{A}_{i_0i_1} \right| \\ &= \frac{X_1 - X_0}{d_{i_0i_1}} \end{aligned} \quad (67)$$

where $\vec{n}_{i_0i_1}$ is the unit normal of $L_{i_0i_1}$, and $d_{i_0i_1}$ is the Euclidean length of edge $i_0 \rightarrow i_1$ belonging to triangle (i_0, i_1, i_2) . The total-flux expression for control-volume i is:

$$F_T = \sum_{k=1}^5 \frac{X_k - X_0}{d_{i_0i_k}} L_{i_0i_k} \quad (68)$$

Thus for gradient-driven fluxes, only the gradient projection onto each edge needs evaluation; full vector direction is not required. This formulation also applies to other fluxes linearly proportional to gradients.

3.3 Flux Evaluation Using Element-Edge Finite-Volume Method

3.3.1 Definition of Element Edge

Conventional flux-evaluation works well for linear dielectric media but fails for nonlinear media such as ferroelectrics. Even the simplest ferroelectric polarization response requires evaluation of vector components along three mutually-orthogonal spatial directions. To correctly simulate ferroelectric polarization, one must compute the true vector orientation of the electric field on each Edge-Couple and evaluate polarization components projected onto individual coordinate axes.

This work adopts the ****Element-Edge (Coupled-Element-Edge)**** method originally proposed by Laux et al. [60,61] for impact-ionization calculations³. The Element-Edge scheme fur-

³No standard Chinese terminology was found for this concept; the term “Element Edge” is coined in this document based on the English name “Element Edge” and the dual-partition property of finite-volume discretization.

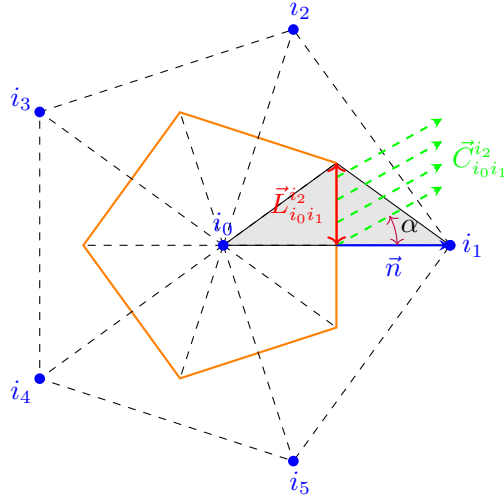


Figure 13: Diagram for element edge method for flux evaluation in FVM.

ther subdivides the original control-volume flux-evaluation domain to evaluate gradient-driven fluxes on finer geometric sub-entities; see Figure 13.

Total flux of field $\vec{C}$ outward across the control-volume boundary of node i_0 in Figure 13 becomes:

$$F_T = \sum_{k=1}^5 \left[\vec{C}_{i_0 i_k}^{i_2} \cdot \vec{L}_{i_0 i_k}^{i_2} + \vec{C}_{i_0 i_k}^{i_5} \cdot \vec{L}_{i_0 i_k}^{i_5} \right] \quad (69)$$

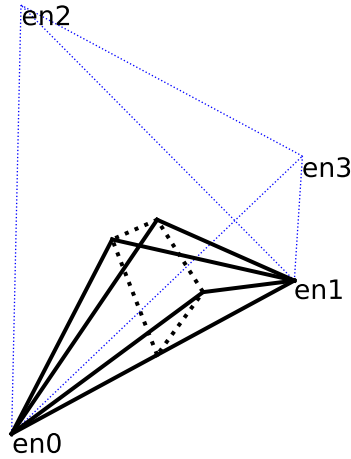


Figure 14: The element edge of one Element cell in 3-D.

$\vec{L}_{i_0 i_k}^{i_2}$ and $\vec{L}_{i_0 i_k}^{i_5}$ denote Element-Edge geometric entities associated with nodes i_0 and i_1 . In two-dimensional meshes each physical edge belongs to two triangular elements, so Element-Edges always appear in pairs. In three dimensions, Element-Edges are partial segments of control-volume outer surfaces as shown by dashed lines in Figure 14. The summation in Equation (69) runs over all tetrahedral elements sharing the physical edge $D_{i_0 i_k}$.

3.3.2 Assembly of Discrete Equations for Newton-Raphson Solver

The Newton-Raphson iterative scheme is commonly adopted for solving finite-volume discretized systems [57, 58]. The iterative update relation for partial-differential systems reads⁴:

$$\mathcal{J}\Delta\mathcal{X} = -\Delta\mathcal{R} \quad (70)$$

where $\mathcal{J}$ is the Jacobian matrix of the residual system, $\Delta\mathcal{X}$ is the solution-space update vector, and $\Delta\mathcal{R}$ is the residual update vector. As iterations proceed, $|\Delta\mathcal{R}| \rightarrow 0$ and $|\Delta\mathcal{X}| \rightarrow 0$, and $\mathcal{X}$ converges toward the physical solution.

After introducing Element-Edge entities, the continuity-equation form for each control volume becomes:

$$\nabla \cdot \vec{C} + \nabla \cdot \vec{A} + G = 0 \quad (71)$$

where $\vec{C}$ represents fluxes evaluated via the Element-Edge formalism, $\vec{A}$ represents conventionally-computed fluxes, and G denotes net volumetric generation rate.

Combining Equation (66) and Equation (69), the discrete residual for Newton iteration reads:

$$\sum_{k,l} \vec{C}_{i_0 i_k}^{i_l} \cdot \vec{L}_{i_0 i_k}^{i_l} + \sum_k \vec{A}_{i_0 i_k} \cdot \vec{L}_{i_0 i_k} + B_i V_i = R_i \quad (72)$$

Here index k enumerates all nodes connected to node i by physical edges; index l enumerates Element-Edge sub-entities associated with $\vec{L}_{i_0 i_k}$; V_i denotes node volume. The first summation term corresponds to Element-Edge-computed flux; the second summation corresponds to conventional gradient-based flux; the final term accounts for net generation within the control volume. Splitting total flux into these two parts enables mixing Element-Edge evaluations with standard finite-volume gradient-flux calculations, lowering computational complexity and improving runtime efficiency.

The corresponding Jacobian-matrix entry is:

$$J_{ij} = \sum_{k,l} \frac{\partial \vec{C}_{i_0 i_k}^{i_l}}{\partial X_j} \cdot \vec{L}_{i_0 i_k}^{i_l} + \sum_k \frac{\partial \vec{A}_{i_0 i_k}}{\partial X_j} \cdot \vec{L}_{i_0 i_k} + \frac{\partial B_i}{\partial X_j} V_i \quad (73)$$

For multi-equation semiconductor simulations with multiple continuity equations, one extends these equations and solution vector $\mathbf{X}$ accordingly. Equation (73) highlights that Newton-Raphson solving requires accurate partial derivatives of residual terms with respect to solution variables. Subsequent sections explicitly list required partial-derivative relations.

3.3.3 Gradient-Vector Evaluation on Element-Edges

To simulate nonlinear ferroelectric polarization, one requires accurate evaluation of electric-field vectors, which are defined as the negative gradient of the solved potential variable. Quantities defined as first-order spatial derivatives of solution unknowns are termed gradient vectors. Below we describe gradient-vector $\vec{C}$ evaluation on Element-Edges for three-dimensional setups.

⁴Calligraphic uppercase symbols denote functions or vectors in solution space.

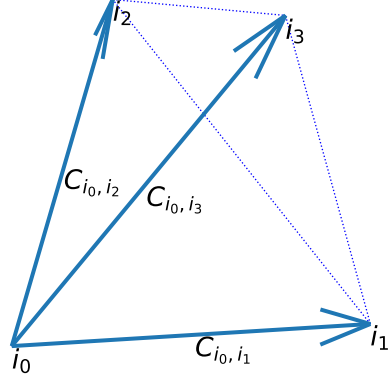


Figure 15: The projects of vector $\vec{C}$ on the edges of one Element cell in 3-D.

From Equation (67), the projection of gradient vector $\vec{C}$ onto physical edge $d_{i_0 i_k}$ connected to node i reads:

$$C_{i_0 i_k} = \vec{C}_i \cdot \vec{n}_{i_0 i_k} = \frac{X_k - X_i}{d_{i_0 i_k}} \quad (74)$$

where $\vec{C}_i$ is the vector value inside tetrahedral element i . Partial derivatives of this projection with respect to nodal solution unknowns are:

$$\frac{\partial C_{i_0 i_k}}{\partial X_i} = -\frac{1}{d_{i_0 i_k}} \quad (75a)$$

$$\frac{\partial C_{i_0 i_k}}{\partial X_k} = \frac{1}{d_{i_0 i_k}} \quad (75b)$$

Figure 15 illustrates vector projections onto edges inside a tetrahedral element. By vector algebra:

$$C_{i_0 i_k} = S_{i_0 i_k} \cdot \vec{C}_i \quad (76)$$

where $S_{i_0 i_k}$ stores unit-edge-vector information for edges inside the element. Written in matrix form for three edges inside one tetrahedron:

$$S_i C_i = \begin{pmatrix} C_{i_0 i_1} \\ C_{i_0 i_2} \\ C_{i_0 i_3} \end{pmatrix} \quad (77)$$

where matrix S_i contains unit-edge vectors originating from node i_0 :

$$S_i = \begin{pmatrix} s_{0,1}^x & s_{0,1}^y & s_{0,1}^z \\ s_{0,2}^x & s_{0,2}^y & s_{0,2}^z \\ s_{0,3}^x & s_{0,3}^y & s_{0,3}^z \end{pmatrix} \quad (78)$$

and vector C_i stores the three-component unknown vector at node i :

$$C_i = \begin{pmatrix} C_i^x \\ C_i^y \\ C_i^z \end{pmatrix} \quad (79)$$

Solving this linear system yields the vector $\vec{C}$ inside each tetrahedral element.

Since Equation (77) involves four nodal unknowns, corresponding Jacobian relations are:

$$S_i \frac{\partial C_i}{\partial X_0} = \left(\frac{\partial C_{i_0 i_1}}{\partial X_0} \quad \frac{\partial C_{i_0 i_2}}{\partial X_0} \quad \frac{\partial C_{i_0 i_3}}{\partial X_0} \right)^T \quad (80a)$$

$$S_i \frac{\partial C_i}{\partial X_1} = \left(\frac{\partial C_{i_0 i_1}}{\partial X_1} \quad 0 \quad 0 \right)^T \quad (80b)$$

$$S_i \frac{\partial C_i}{\partial X_2} = \left(0 \quad \frac{\partial C_{i_0 i_2}}{\partial X_2} \quad 0 \right)^T \quad (80c)$$

$$S_i \frac{\partial C_i}{\partial X_3} = \left(0 \quad 0 \quad \frac{\partial C_{i_0 i_3}}{\partial X_3} \right)^T \quad (80d)$$

Because each physical edge connects two nodes inside a tetrahedral mesh, one evaluates the average vector value over the two adjacent nodes i and j for flux evaluation on Element-Edge faces:

$$\vec{C}_{ij} = \frac{1}{2} (C_i + C_j) \quad (81)$$

Associated Jacobian:

$$\frac{\partial \vec{C}_{ij}}{\partial X_k} = \frac{1}{2} \left(\frac{\partial C_i}{\partial X_k} + \frac{\partial C_j}{\partial X_k} \right) \quad (82)$$

Index k denotes arbitrary node illustrated in Figure 15.

In finite-volume discretization, flux leaving one control volume must enter its neighbor. Taking the nodal average as in Equation (81) ensures inter-control-volume consistency. Since S_i depends purely on mesh geometry, it needs precomputing only once.

After obtaining $\vec{C}$ and its derivatives at tetrahedral-element nodes from Equation (81) and Equation (82), one substitutes these quantities into Equation (72) and Equation (73) to assemble discrete residual equations for iterative solving.

3.3.4 Practical Remarks on Element-Edge Computation

The preceding subsection computes vector $\vec{C}$ inside a tetrahedron using edge projections evaluated at one node. The index notation in Equation (81) is more generic and can be reused for other discretization schemes; additional indices in Equation (72) tag which tetrahedral element the flux-vector evaluation belongs to. Therefore a single physical edge yields multiple flux-vector evaluations corresponding to each attached tetrahedral element (two evaluations in two-dimensional meshes).

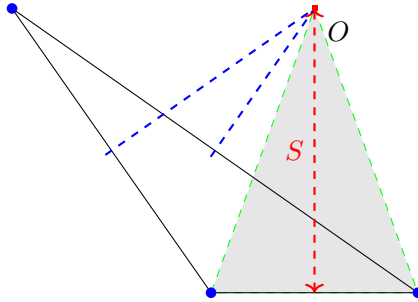


Figure 16: The diagram of element edge for obtuse triangle.

Robust convergence requires triangular or tetrahedral mesh elements **free of obtuse interior angles**. For obtuse triangles, the circumcenter lies outside the element domain (see Figure 16), and fluxes evaluated via Element-Edge entities overestimate physical values, causing numerical divergence. Generating purely acute meshes via circumcenter-based dual partitioning is non-trivial. When convergence difficulties arise, mesh refinement techniques described in Section ?? can improve mesh quality.

Since Newton-Raphson iteration is employed, accurate Jacobian-matrix entries are critical for numerical convergence; the correctness of all partial-derivative relations directly governs whether simulations converge.

3.4 Assembly of Semiconductor Device Equations

3.4.1 Electron and Hole Continuity Equations

In semiconductor-device simulation, electron and hole concentrations represent two conserved physical quantities. Combining Equation (16) and Equation (62), the continuity equations for electrons and holes read:

$$\frac{\partial n}{\partial t} = \nabla \cdot (n\mu_n \vec{E} + D_n \nabla n) + G_n \quad (83a)$$

$$\frac{\partial p}{\partial t} = \nabla \cdot (-p\mu_p \vec{E} + D_p \nabla p) + G_p \quad (83b)$$

Particle motion splits into electric-field-driven drift contributions and thermally-driven diffusion contributions; these are collectively known as drift-diffusion equations.

One major distinction between semiconductor simulation and conventional computational fluid dynamics: carrier concentrations in semiconductors, governed by Fermi statistics, can vary by many orders-of-magnitude within tiny spatial regions. This effect is directly observable experimentally, e.g. on-off ratios exceeding five orders-of-magnitude in transistor transfer characteristics. Although CFD methodologies transfer readily to thermal-flow, phase-change and particle-transport problems, special modifications are required for semiconductor carrier transport.

3.4.2 Gauss' s Law for Electrostatics

Magnetic-field effects are usually neglected for device-level electrostatics simulations. One solves for electrostatic scalar potential, reducing full 3-D vector-field computation to scalar-field computation. Gauss' s law for electrostatics yields a time-independent conservation relation:

$$\nabla^2 \varphi = -\frac{q}{\varepsilon_0 \varepsilon_r} \quad (84)$$

where $\vec{E}$ is electric-field intensity, q is volumetric charge density, ε_0 is vacuum permittivity, and ε_r denotes material relative permittivity.

For ordinary linear dielectrics with simple cubic lattices, ε_r is constant. For ferroelectric materials, however, the effective permittivity becomes dependent upon polarization history. Although one could treat ferroelectrics using differential permittivity as defined in Equation

(60), this approach carries intrinsic drawbacks (Section 2.4.2). This work directly employs the electric-displacement-vector formulation including explicit ferroelectric polarization:

$$\nabla \cdot \vec{D} = \nabla \cdot (\varepsilon_0 \varepsilon_r \vec{E} + \vec{P}) = q \quad (85)$$

Here ε_r represents the paraelectric-phase background relative permittivity of ferroelectric material, and polarization $\vec{P}$ is computed either from Equation (56) or Equation (57).

3.4.3 Scharfetter-Gummel Discretization

Equation (83) shows carrier flux contains both drift and diffusion terms. Straightforward discretization suffers from “large-number swamping small-number” numerical artifacts. In addition, semiconductor devices feature extremely steep carrier-concentration gradients in critical device regions, where conventional discretization performs poorly. The Scharfetter-Gummel discretization scheme provides an established remedy [62,63]. We briefly illustrate it for one-dimensional cases.

Rewrite the current-density expression (16) in generic particle-flux form:

$$j = -D \frac{\partial n}{\partial x} + vn \quad (86)$$

where j is particle-number flux density, D is diffusion coefficient, n particle concentration, x spatial coordinate, v drift velocity. Considering only flux contributions to concentration evolution, the time derivative at node i becomes:

$$\frac{\partial n_i}{\partial t} = -\frac{j_{i+1} - j_{i-1}}{\Delta} \quad (87)$$

where j_{i+1} , j_{i-1} are flux densities across adjacent cell interfaces, and Δ represents effective width associated with node i . Concentrations at neighboring grid points are n_i , n_{i+1} . Flux exists across both interfaces adjacent to node i .

Every first-order linear ordinary differential equation admits an integrating factor. Multiply the flux j_{i+1} between node i and $i+1$ by the integrating factor:

$$\begin{aligned} j_{i+1} \exp\left(-v \frac{x - x_i}{D}\right) &= \left(-D \frac{\partial n}{\partial x} + vn\right) \exp\left(-v \frac{x - x_i}{D}\right) \\ &= -D \frac{\partial}{\partial x} \left[n \exp\left(-v \frac{x - x_i}{D}\right) \right] \end{aligned} \quad (88)$$

x is spatial coordinate; x_i , x_{i+1} denote coordinates for node n_i and n_{i+1} . Integrate over interval $[x_i, x_{i+1}]$:

$$\begin{aligned} \int_{x_i}^{x_{i+1}} j_{i+1} \exp\left(-v \frac{x - x_i}{D}\right) dx &= \int_{x_i}^{x_{i+1}} -D \frac{\partial}{\partial x} \left[n \exp\left(-v \frac{x - x_i}{D}\right) \right] dx \\ j_{i+1} \frac{D}{v} \exp\left(-v \frac{d}{D}\right) &= D \left[n_i - n_{i+1} \exp\left(-v \frac{d}{D}\right) \right] \end{aligned} \quad (89)$$

where $d = |x_{i+1} - x_i|$. Final simplification yields:

$$j_{i+1} = v \frac{n_i - n_{i+1} \exp\left(-v \frac{d}{D}\right)}{1 - \exp\left(-v \frac{d}{D}\right)} \quad (90)$$

Comparing Equation (86) and Equation (90), the Scharfetter-Gummel scheme condenses drift-diffusion flux into a single expression. It suppresses precision loss originating from “large-number swamping small-number” arithmetic [64]. This substantially improves finite-volume-method applicability for semiconductor simulation and enables robust numerical treatment of steep concentration gradients inside depletion and accumulation layers.

3.4.4 Semiconductor-Electrode Interfaces —Boundary Conditions

Equation (83) and Equation (84) describe transport laws valid inside homogeneous material domains. Boundary conditions must be specified for quantities at domain interfaces (e.g. carrier injection at contacts). Numerical boundary conditions are conventionally classified into two families:

- a. **Dirichlet boundary condition (first-type boundary condition):** Prescribe the value of unknowns directly on domain boundaries.

For semiconductor devices, Dirichlet conditions assign carrier concentrations and electrostatic potentials on contact boundaries for Equation (83) and Equation (84). Potentials at electrodes are directly fixed by external bias, so contacts typically employ Dirichlet conditions. Two cases exist for carrier boundary values. Ohmic contacts feature zero interfacial potential barrier; heavily-doped silicon electrodes provide a typical example. Boundary carrier concentrations are determined from charge neutrality and doping concentration, which also corresponds to Dirichlet boundary conditions. For inorganic crystalline semiconductors with intrinsic carrier concentration n_i :

$$\begin{cases} p - n + N_{net} &= 0 \\ pn &= n_i^2 \end{cases} \quad (91)$$

where N_{net} is net doping concentration. Solving yields:

$$n = n_i + \frac{1}{2} \left(\sqrt{N_{net}^2 + 4n_i^2} + N_{net} \right) \quad (92a)$$

$$p = n_i + \frac{1}{2} \left(\sqrt{N_{net}^2 + 4n_i^2} - N_{net} \right) \quad (92b)$$

- b. **Neumann boundary condition (second-type boundary condition):** Prescribe first-order spatial derivatives of unknown quantities on boundaries.

Schottky contacts formed at metal-semiconductor junctions feature an interfacial potential barrier [45]. Current density is computed from thermionic-emission theory [2], which fits naturally into Neumann-type boundary conditions. This simulation adopts the simplified formulation used in Synopsys Sdevice [65]:

$$\vec{J}_n \cdot \hat{n} = ev_p(n - n_0) \quad (93a)$$

$$\vec{J}_p \cdot \hat{n} = ev_p(p - p_0) \quad (93b)$$

where $\hat{n}$ denotes outward-pointing interface normal; n_0, p_0 are equilibrium carrier concentrations. Thermal emission velocities for electrons and holes read:

$$v_i = \sqrt{\frac{k_B T}{2m_i^* \pi}} \quad (94)$$

where m_i^* is carrier effective mass.

References

- [1] 黄昆. 固体物理. 北京大学出版社, 北京, 2014.
- [2] Donald A. Neamen. 半导体物理与器件. 第四版. 电子工业出版社, 北京, 2013.
- [3] D. M. Caughey and R. E. Thomas. Carrier mobilities in silicon empirically related to doping and field. *Proceedings of the IEEE*, 55(12):2192–2193, 1967.
- [4] N. D. Arora, J. R. Hauser, and D. J. Roulston. Electron and hole mobilities in silicon as a function of concentration and temperature. *IEEE Transactions on Electron Devices*, 29(2):292–295, 1982.
- [5] R. H. Friend, R. W. Gymer, A. B. Holmes, J. H. Burroughes, R. N. Marks, C. Taliani, D. D. C. Bradley, D. A. Dos Santos, J. L. Bredas, M. Logdlund, and W. R. Salaneck. Electroluminescence in conjugated polymers. *Nature*, 397(6715):121–128, 1999.
- [6] M Pfeiffer, K Leo, X Zhou, J.S Huang, M Hofmann, A Werner, and J Blochwitz-Nimoth. Doped organic semiconductors: Physics and application in light emitting diodes. *Organic Electronics*, 4(2):89–103, 2003. High Efficiency Light Emitters.
- [7] A. Taha, A.A.M. Farag, O.M.I. Adly, N. Roushdy, Magdy Shebl, and H.M. Ahmed. Photoresponse and dft studies of new synthesized 2-benzylidene-3-hydroxy -1-(5,6-diphenyl-1,2,4-triazine-3-yl)hydrazine and optical sensor application. *Journal of Molecular Structure*, 1142:66–72, 2017.
- [8] Sergei D. Baranovskii. Mott lecture: Description of charge transport in disordered organic semiconductors: Analytical theories and computer simulations. *physica status solidi (a)*, 215(12):1700676, 2018.
- [9] J. O. Oelerich, D. Huemmer, and S. D. Baranovskii. How to find out the density of states in disordered organic semiconductors. *Physical Review Letters*, 108(22):226403, 2012.
- [10] E. Knapp, R. Häusermann, H. U. Schwarzenbach, and B. Ruhstaller. Numerical simulation of charge transport in disordered organic semiconductor devices. *Journal of Applied Physics*, 108(5):054504, 2010.
- [11] J. Li, Y. Zhao, H. S. Tan, Y. Guo, C. A. Di, G. Yu, Y. Liu, M. Lin, S. H. Lim, Y. Zhou, H. Su, and B. S. Ong. A stable solution-processed polymer semiconductor with record high-mobility for printed transistors. *Scientific Reports*, 2:754, 2012.
- [12] Vladimir S. Bystrov, Ekaterina V. Paramonova, Yuri Dekhtyar, Robert C. Pullar, Aleksey Katashev, Natalie Polyaka, Anna V. Bystrova, Alla V. Saprionova, Vladimir M. Fridkin, Herbert Kliem, and Andrei L. Kholkin. Polarization of poly(vinylidene fluoride) and poly(vinylidene fluoride-trifluoroethylene) thin films revealed by emission spectroscopy with computational simulation during phase transition. *Journal of Applied Physics*, 111(10), 2012.

- [13] S. L. M. van Mensfoort and R. Coehoorn. Effect of gaussian disorder on the voltage dependence of the current density in sandwich-type devices based on organic semiconductors. *Physical Review B*, 78(8), 2008.
- [14] Gernot Paasch and Susanne Scheinert. Charge carrier density of organics with gaussian density of states: Analytical approximation for the gauss–fermi integral. *Journal of Applied Physics*, 107(10), 2010.
- [15] Jerry P. Selvaggi. Analytical evaluation of the charge carrier density of organic materials with a gaussian density of states revisited. *Journal of Computational Electronics*, 17(1):61–67, 2017.
- [16] S. L. M. van Mensfoort, S. I. E. Vulto, R. A. J. Janssen, and R. Coehoorn. Hole transport in polyfluorene-based sandwich-type devices: Quantitative analysis of the role of energetic disorder. *Physical Review B*, 78(8), 2008.
- [17] S. W. Tsang, S. K. So, and J. B. Xu. Application of admittance spectroscopy to evaluate carrier mobility in organic charge transport materials. *Journal of Applied Physics*, 99(1):013706, 2006.
- [18] J. L. Hartke. The three-dimensional poole-frenkel effect. *Journal of Applied Physics*, 39(10):4871–4873, 1968.
- [19] R. Coehoorn, W. F. Pasveer, P. A. Bobbert, and M. A. J. Michels. Charge-carrier concentration dependence of the hopping mobility in organic materials with gaussian disorder. *Physical Review B*, 72(15), 2005.
- [20] M. Ghittorelli, T. Lenz, H. Sharifi Dehsari, D. Zhao, K. Asadi, P. W. M. Blom, Z. M. Kovacs-Vajna, D. M. de Leeuw, and F. Torricelli. Quantum tunnelling and charge accumulation in organic ferroelectric memory diodes. *Nature Communications*, 8(1):15741, 2017.
- [21] N. F. Mott. Conduction in non-crystalline materials. *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics*, 19(160):835–852, 1969.
- [22] Yongjeong Lee, Sungyeop Jung, Andrew Plews, Ahmed Nejim, Olivier Simonetti, Louis Giraudet, Sergei D. Baranovskii, Florian Gebhard, Klaus Meerholz, Sungjune Jung, Gilles Horowitz, and Yvan Bonnasieux. Parametrization of the gaussian disorder model to account for the high carrier mobility in disordered organic transistors. *Physical Review Applied*, 15(2):024021, 2021.
- [23] Hamna F. Haneef, Andrew M. Zeidell, and Oana D. Jurchescu. Charge carrier traps in organic semiconductors: a review on the underlying physics and impact on electronic devices. *Journal of Materials Chemistry C*, 8(3):759–787, 2020.
- [24] H. T. Nicolai, M. M. Mandoc, and P. W. M. Blom. Electron traps in semiconducting polymers: Exponential versus gaussian trap distribution. *Physical Review B*, 83:195204, May 2011.

- [25] José M. Montero and Juan Bisquert. Interpretation of trap-limited mobility in space-charge limited current in organic layers with exponential density of traps. *Journal of Applied Physics*, 110(4):043705, 2011.
- [26] Mutsumi Kimura, Takashi Nakanishi, Kenji Nomura, Toshio Kamiya, and Hideo Hosono. Trap densities in amorphous-ingazno4 thin-film transistors. *Applied Physics Letters*, 92(13):133512, 2008.
- [27] Hui Xu, Wen-Juan Zhai, Chao Tang, Shao-Ya Qiu, Rui-Lan Liu, Zhou Rong, Zong-Qiang Pang, Bing Jiang, Jing Xiao, Chao Zhong, Bao-Xiu Mi, Qu-Li Fan, and Wei Huang. Thickness dependence of carrier mobility and the interface trap free energy investigated by impedance spectroscopy in organic semiconductors. *The Journal of Physical Chemistry C*, 120(31):17184–17189, 2016.
- [28] Patricio Farrell, Thomas Koprucki, and Jürgen Fuhrmann. Computational and analytical comparison of flux discretizations for the semiconductor device equations beyond boltzmann statistics. *Journal of Computational Physics*, 346:497–513, 2017.
- [29] Gernot Paasch and Susanne Scheinert. Charge carrier density of organics with gaussian density of states: Analytical approximation for the gauss–fermi integral. *Journal of Applied Physics*, 107(10):104501, 2010.
- [30] A. Boubaker, B. Hafsi, K. Lmimouni, and A. Kalboussi. A comparative tcad simulations of a p-and n-type organic field effect transistors: field-dependent mobility, bulk and interface traps models. *Journal of Materials Science: Materials in Electronics*, 28(11):7834–7843, 2017.
- [31] Dirk M. Guldi, Chuping Luo, Maurizio Prato, Elke Dietel, and Andreas Hirsch. Charge-transfer in a π -stacked fullerene porphyrin dyad: evidence for back electron transfer in the ‘marcus-inverted’ region. *Chemical Communications*, pages 373–374, 2000.
- [32] A. Hepp, H. Heil, W. Weise, M. Ahles, R. Schmechel, and H. von Seggern. Light-emitting field-effect transistor based on a tetracene thin film. *Physical Review Letters*, 91(15):157406, 2003. Hepp, Aline Heil, Holger Weise, Wieland Ahles, Marcus Schmechel, Roland von Seggern, Heinz eng Phys Rev Lett. 2003 Oct 10;91(15):157406. doi: 10.1103/Phys-RevLett.91.157406. Epub 2003 Oct 10.
- [33] Allen Miller and Elihu Abrahams. Impurity conduction at low concentrations. *Physical Review*, 120(3):745–755, 1960.
- [34] A. Yelon, B. Movaghar, and R. S. Crandall. Multi-excitation entropy: its role in thermodynamics and kinetics. *Reports on Progress in Physics*, 69(4):1145–1194, 2006.
- [35] Z. G. Yu, D. L. Smith, A. Saxena, R. L. Martin, and A. R. Bishop. Molecular geometry fluctuations and field-dependent mobility in conjugated polymers. *Physical Review B*, 63(8), 2001.

- [36] M. V. Basilevsky, A. V. Odínokov, and K. G. Komarova. Charge-transfer mobility parameters in photoelectronic devices: The advanced miller-abrahams computation. *J Phys Chem B*, 119(24):7430–8, 2015. Basilevsky, M V Odínokov, A V Komarova, K G eng J Phys Chem B. 2015 Jun 18;119(24):7430-8. doi: 10.1021/jp5110714. Epub 2015 Jan 30.
- [37] N. J. van der Kaap, I. Katsouras, K. Asadi, P. W. M. Blom, L. J. A. Koster, and D. M. de Leeuw. Charge transport in disordered semiconducting polymers driven by nuclear tunneling. *Physical Review B*, 93(14), 2016.
- [38] M. Scheb, C. Zimmermann, and C. Jungemann. Field-induced detrapping in doped organic semiconductors with gaussian disorder and different carrier localizations on host and guest sites. *Physical Review B*, 92:104201, Sep 2015.
- [39] Davood Abbaszadeh, Alexander Kunz, Naresh B. Kotadiya, Anirban Mondal, Denis Andrienko, Jasper J. Michels, Gert-Jan A. H. Wetzelaer, and Paul W. M. Blom. Electron trapping in conjugated polymers. *Chemistry of Materials*, 31(17):6380–6386, 2019.
- [40] N. J. Harmon and M. E. Flatté. Spin-flip induced magnetoresistance in positionally disordered organic solids. *Physical Review Letters*, 108(18), 2012.
- [41] N. J. Harmon and M. E. Flatté. Semiclassical theory of magnetoresistance in positionally disordered organic semiconductors. *Physical Review B*, 85(7), 2012.
- [42] P. Bobbert, W. Wagemans, F. van Oost, B. Koopmans, and M. Wohlgenannt. Theory for spin diffusion in disordered organic semiconductors. *Physical Review Letters*, 102(15), 2009.
- [43] P. Bobbert, T. Nguyen, F. van Oost, B. Koopmans, and M. Wohlgenannt. Bipolaron mechanism for organic magnetoresistance. *Physical Review Letters*, 99(21), 2007.
- [44] Yogesh Yadav and Samarendra Pratap Singh. Effect of dielectric surface passivation on organic field-effect transistors: spectral analysis of the density of trap-states. *Semiconductor Science and Technology*, 37(1):015015, 2021.
- [45] Kwok K. Ng S.M. Sze. *Chapter 4 - Metal-Insulator-Semiconductor Capacitors*, pages 197–240. 2006.
- [46] Peter Maksymovych, Jan Seidel, Ying Hao Chu, Pingping Wu, Arthur P. Baddorf, Long-Qing Chen, Sergei V. Kalinin, and Ramamoorthy Ramesh. Dynamic conductivity of ferroelectric domain walls in bifeo₃. *Nano Letters*, 11(5):1906–1912, 2011.
- [47] Mengwei Si, Atanu K. Saha, Shengjie Gao, Gang Qiu, Jingkai Qin, Yuqin Duan, Jie Jian, Chang Niu, Haiyan Wang, Wenzhuo Wu, Sumeet K. Gupta, and Peide D. Ye. A ferroelectric semiconductor field-effect transistor. *Nature Electronics*, 2(12):580–586, 2019.
- [48] S. L. Miller and P. J. McWhorter. Physics of the ferroelectric nonvolatile memory field effect transistor. *Journal of Applied Physics*, 72(12):5999–6010, 1992.
- [49] VL Ginzburg. On the dielectric properties of ferroelectric (segnette-electric) crystals and barium titanate. *Journal of Experimental and Theoretical Physics*, 15:1945, 1945.

- [50] 钟维烈. 铁电体物理学. 凝聚态物理学丛书. 科学出版社, 北京, 1996.
- [51] Biao Wang, R. Xia, H. Fan, and C. H. Woo. Dynamic process of domain switching in ferroelectric films. *Journal of Applied Physics*, 94(5):3384–3389, 2003.
- [52] Binjian Zeng, Chen Liu, Siwei Dai, Pingan Zhou, Keyu Bao, Shuaizhi Zheng, Qiangxiang Peng, Jinjuan Xiang, Jianfeng Gao, Jie Zhao, Min Liao, and Yichun Zhou. Electric field gradient-controlled domain switching for size effect-resistant multilevel operations in hfo 2-based ferroelectric field-effect transistor. *Advanced Functional Materials*, 31(17), 2021.
- [53] T.Kamei, Eisuke Tokumitsu, and Hiroshi Ishiwara. Numerical analysis of metal-ferroelectric- semiconductor field effect transistors (mfs-fets) considering inhomogeneous ferroelectric polarization. *IEICE Transactions on Electronics*, E81-C:577–583, 1998.
- [54] Yong C. Gan and Wan Q. Cao. Mechanism analyses of differential and static dielectric constants in ferroelectrics. *Ferroelectrics Letters Section*, 47(4-6):96–104, 2020.
- [55] N. Bar-Chaim, M. Brunstein, J. Grünberg, and A. Seidman. Electric field dependence of the dielectric constant of pzt ferroelectric ceramics. *Journal of Applied Physics*, 45(6):2398–2405, 1974.
- [56] Vladimir O. Sherman, Alexander K. Tagantsev, Nava Setter, David Iddles, and Tim Price. Ferroelectric-dielectric tunable composites. *Journal of Applied Physics*, 99(7):074104, 2006.
- [57] 李荣华. 偏微分方程数值解法. 高等教育出版社, 北京, 2005.
- [58] Siegfried Selberherr. *Analysis and simulation of semiconductor devices*. Springer-Verlag, NY, 1984.
- [59] 唐俭姜健飞, 胡良剑. 数值分析及其 *MATLAB* 实验. 科学出版社, 北京, 2004.
- [60] Steven E. Laux and Robert G. Byrnes. Semiconductor device simulation using generalized mobility models. *IBM Journal of Research and Development*, 29(3):289–301, May 1985.
- [61] S. E. Laux and B. M. Grossman. A general control-volume formulation for modeling impact ionization in semiconductor transport. *IEEE Trans Electron Devices*, 32(10):2076–2082, 1985.
- [62] D. L. Scharfetter and H. K. Gummel. Large-signal analysis of a silicon read diode oscillator. *IEEE Transactions on Electron Devices*, 16(1):64–77, 1969.
- [63] William R. Frensley. Scharfetter-gummel discretization scheme for drift-diffusion equations, 2004.
- [64] Marianne Bessemoulin-Chatard. A finite volume scheme for convectiondiffusion equations with nonlinear diffusion derived from the scharfettergummel scheme. *Numerische Mathematik*, 121(4):637–670, 2012.
- [65] *Sentaurus Device User Guide*, chapter 10, page 219. Synopsys, Inc., 2013.